

Redundancy Resolution of Rehabilitative Multi-Segment Gough-Stewart Platforms using Genetic Algorithm

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Abstract—Gough-Stewart platforms (GSPs) are utilized for upper and lower extremity rehabilitation, among other extensive applications. Serially connecting these parallel manipulators opens up novel applications in biomimetics and exoskeletons. In a hybrid two-segment serially connected Gough-Stewart manipulator, twelve actuators control the end-effector's six degrees of freedom. Thus, there are non-unique configurations possible for the robot to reach a desired goal. Known as actuator redundancy, this can serve several objectives, such as obstacle and singularity avoidance. This paper proposes an optimization algorithm using genetic algorithm that resolves the redundancy of the robot for minimum power consumption. The redundancy problem is formulated as a single-objective bound unconstrained problem. A genetic algorithm is implemented in MATLAB® and validated against alternative optimizers including Particle Swarm Optimization and Simulated Annealing. Over adequate test cases, the obtained results demonstrate the algorithm's ability to reach optimal robot configurations. This can be used as an inverse kinematic solution for bio-inspired actuators and rehabilitative exoskeletons.

Index Terms—Gough-Stewart manipulator, optimization, redundancy, genetic algorithm, rehabilitation.

I. INTRODUCTION

The Gough-Stewart Platform (GSP) is a foundational six-degree-of-freedom (6-DOF) parallel manipulator consisting of a top mobile platform connected to a stationary base via six independent, extensible prismatic actuators [1], [2]. Characterized by closed kinematic chains, GSPs offer exceptional structural rigidity, high stiffness, superior positioning accuracy, and a remarkably high load-to-weight ratio [3]. Consequently, GSPs have long been established in classic industrial and aerospace applications [1], [2]. However, standard parallel manipulators suffer from a relatively constrained operational workspace and restricted tilt angles limited by kinematic singularities [4]. To bridge the gap between the superior load capacity and rigidity of parallel configurations and the extensive workspace, reach, and dexterity of serial manipulators, hybrid configurations created by serially connecting multiple Gough-Stewart platforms (SCGPs) have been introduced [3], [5].

In biomedical engineering, single-segment GSPs have been adopted for specialized dynamic testing and single-joint therapy. For posture and balance evaluation, computerized motion platforms such as the Computer Assisted Rehabilitation Environment (CAREN) [6] and the Dynamic Oriented Rehabilitative Integrated System (DORIS) [7] utilize 6-DOF parallel bases to apply dynamic perturbations during posturography and gait analysis. Similar hexapod footplate mechanisms have been engineered as bipedal gait test systems to simulate ground reaction forces [8], while specialized multi-platform structures—such as the 2(3HUS+S) manipulator—have been built to simulate complex hip joint kinematics during walking cycles [9]. For therapeutic intervention, single 6-DOF parallel orthoses have demonstrated success in executing post-stroke lower limb exercises [10], [11].

While single platforms are effective for isolated foot or ankle motions, multi-segment SCGPs provide the hyper-redundant degrees of freedom required for multi-joint, large-workspace clinical applications [3], [5]. In orthopedics, a prominent clinical manifestation of multi-strut parallel architecture is the Taylor Spatial Frame (TSF), where hexapod fixators are used in serial alignment along bone segments to perform multi-axis fracture reduction, bone lengthening, and severe deformity correction [5], [12]. From a biomimetic perspective, SCGPs naturally replicate complex biological structures—such as the human arm, multi-segment spine, or flexible continuum appendages (e.g., elephant trunks) [3], [5], [13]. Fig. 1 shows the musculoskeletal system of human upper extremity, extracted from OpenSim® model [14]. For simplicity, it only depicts a representative fiber for each of the main muscle groups that control the movement of arm, forearm and wrist. The parallel configuration of muscles around bones in each section, connected serially via elbow joint, highlights the serial-parallel architecture that resembles the SCGP shown in Fig. 2 [5].

When Gough-Stewart modules are cascaded sequentially, the resulting SCGPs exhibit hyper-redundancy, offering flexible multi-joint coordination for biomimetic and rehabilitative applications. This introduces multiple valid joint-space

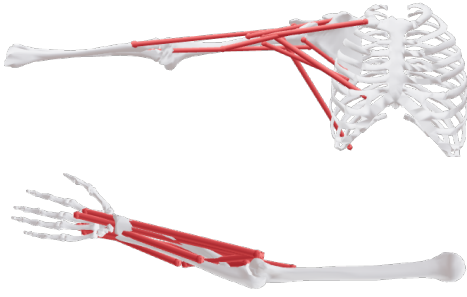


Fig. 1. Human's upper limb musculoskeletal system visualized in OpenSim [14] that shows serial-parallel architecture.

solutions for a single end-effector pose, necessitating robust redundancy resolution algorithms [11]. Existing redundancy resolution methodologies in the literature can be categorized into three distinct paradigms based on the type of redundancy being resolved. In some single-stage parallel mechanisms, kinematic redundancy is incorporated by integrating extra active or passive joints into the leg branches or platform base to reduce mobility constraints. Gosselin and Schreiber [15] developed a kinematically redundant $(6 + 3)$ -DOF Gough-Stewart platform to reorient passive intermediate links and guarantee a singularity-free orientation workspace exceeding tilt angles of 60° .

Unlike kinematic redundancy, actuation redundancy adds extra driven limbs or joints to an existing parallel structure. Boudreau and Léger [16] addressed kinetostatic force allocation in redundantly actuated 3-PRPR planar parallel manipulators, resolving generalized force distribution through a Sequential Quadratic Programming (SQP) optimization algorithm to minimize total actuator driving effort and pass smoothly through singular points. In construction robotics handling heavy loads, Wang et al. [17] developed a Redundantly Actuated Series-Parallel Manipulator (RAS-PM), establishing a Recursive Kinetostatic Optimization Algorithm for hybrid force/position allocation to maximize payload-to-weight ratios.

For hyper-redundant SCGP, Nourian et al. [5] resolved the non-unique inverse kinematics through a generalized Deep Reinforcement Learning architecture based on a recursive Proximal Policy Optimization algorithm optimized for minimal execution time. For discretely actuated Variable Geometry Truss modules, Motahari [18] introduced a Two-By-Two Pending Module Search technique and evaluated its performance against multiple meta-heuristics, explicitly demonstrating that the Genetic Algorithm, Particle Swarm Optimization, and Grey Wolf Optimizer successfully resolve end-effector posture tracking in spatial hyper-redundant chains. For soft continuum bio-inspired manipulators, Huang et al. [13] analyzed a multi-segment series-parallel elephant-trunk robot by using NSGA-II for multi-objective spatial bending trajectory optimization. In rehabilitative parallel robotics, Amini Azar et al. [11] developed a multi-DOF Stewart rehabilitation platform, employing a multidimensional Genetic Algorithm to optimize adaptive impedance control parameters, ensuring smooth therapist-like

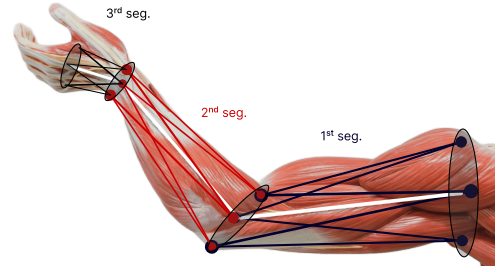


Fig. 2. Representation of human upper limb with a 3 segment serially connected Gough-Stewart platform.

interaction during post-stroke lower-limb gait exercises.

Despite the significant advancements in redundancy resolution strategies, existing approaches have primarily focused on industrial pick-and-place tasks or planar mechanisms, leaving a critical research gap in global search-based redundancy resolution algorithms specifically tailored for multi-segment rehabilitative and biomimetic Gough-Stewart platforms. To bridge this gap, this paper proposes a redundancy resolution framework for a two-segment GSP using Genetic Algorithm. After formulation of the redundancy into an optimization problem, a GA based on a single-objective fitness function is implemented to minimize the gross power consumption of the actuators.

The remainder of this paper is organized as follows: Section II presents the mathematical formulation of the the inverse kinematics and Jacobian derivation. Section III details the optimization objectives and implementation of the Genetic Algorithm. Section IV and V present and discuss simulation results for multiple test cases. Finally, Section VI concludes the paper and outlines potential directions for future research.

II. FORMULATION OF KINEMATICS

Redundancy in actuators means higher number of actuators in a robot than its intended degrees of freedom. A two segment GSP has a total of 12 limbs that help coordinate the 6 degrees of freedom of its overall end-effector. The 6 redundant actuators result in non-unique answers to the Inverse kinematic problem (IKP) of the robot, which refers to obtaining the joint space variables given any specified task-space pose.

A. Inverse Kinematics Problem

As a parallel manipulator, a Gough-Stewart platform has 6 kinematic loops that determine its kinematics. As depicted in Fig. 3, the center of each plate in a GSP has its own frame, notated as O_A for the lower and O_B for the upper plate. Each of the six limbs connects point a_i on the lower plate to point B_i on the upper plate, using a pair of spherical or universal joints. The only actuated joint in each kinematic loop is the prismatic limb connecting the two aforementioned points with the length of l_i which is a function of time. The unit vector along this limb is notated as $\hat{s}_i$. Superscripts A and B in all notations determine the reference frame, and subscript i refers to limb number, ranging from 1 to 6.

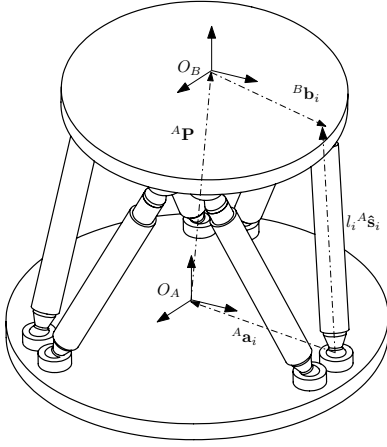


Fig. 3. A Gough-Stewart platform with marked up kinematic loop.

For i^{th} actuator the governing equation can be derived from Fig. 3 as follows [3]:

$$l_i^A \hat{s}_i = {}^A \mathbf{P} + {}^A \mathbf{R}_B {}^B \mathbf{b}_i + {}^A \mathbf{a}_i \quad (1)$$

Vector ${}^A \mathbf{P}$ denotes the position of the end-effector reported in the base frame, i.e. ${}^A \mathbf{O}_B$. Matrix ${}^A \mathbf{R}_B$ is the rotation matrix calculated by the angles θ, ϕ, ψ as the roll, pitch and yaw of the upper plate in frame A . In inverse kinematics, the known pose of the end-effector contains these parameters according to (3).

Among all common shapes like octagon or triangle, this study investigates GSPs with circular plates for both base and end-effector. With the plates' radii known and the angular offset of each spherical joint shown in Fig. 4, geometric parameters ${}^A \mathbf{a}_i$ and ${}^B \mathbf{b}_i$ can be determined within each of their corresponding frames. Taking the Euclidean norm of both sides of (1), one has [19]:

$$l_i^2 = [{}^A \mathbf{P} + {}^A \mathbf{R}_B {}^B \mathbf{b}_i + {}^A \mathbf{a}_i]^T [{}^A \mathbf{P} + {}^A \mathbf{R}_B {}^B \mathbf{b}_i + {}^A \mathbf{a}_i] \quad (2)$$

The transformation matrix for the i -th GS segment, in its own base frame is calculated as [20]:

$${}^{i-1} \mathbf{T}_i = \begin{bmatrix} {}^{i-1} \mathbf{R}_i & {}^{i-1} \mathbf{P}_i \\ 1 \times 3 & 1 \end{bmatrix}_{4 \times 4} \quad (3)$$

where ${}^{i-1} \mathbf{R}_i$ and ${}^{i-1} \mathbf{P}_i$ in this equation stand for the rotation matrix and position vector of the i^{th} GSP segment with respect to its base frame (frame $i - 1$), respectively.

For a hybrid manipulator with N segments, serially multiplying the transformation matrices for all segments results in the transformation matrix of the final end-effector in the global frame [21], [22]:

$${}^G \mathbf{T}_N = {}^G \mathbf{T}_1 {}^1 \mathbf{T}_2 {}^2 \mathbf{T}_3 \dots {}^{N-1} \mathbf{T}_N \quad (4)$$

where G stands for the global frame of the robot. For two GS segments, one can rewrite this equation as:

$${}^1 \mathbf{T}_2 = ({}^G \mathbf{T}_1)^{-1} {}^G \mathbf{T}_2 \quad (5)$$

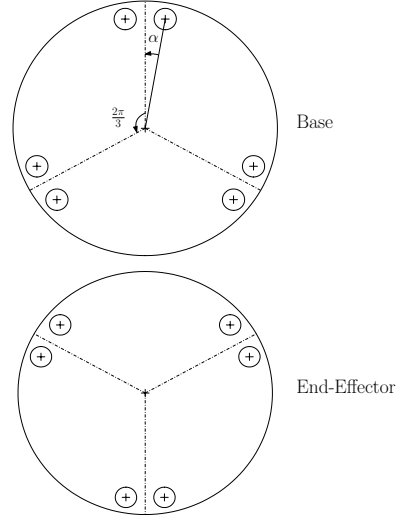


Fig. 4. The angular offset of spherical joints on the base and end-effector of the GSP under study.

which contains the position (${}^1 \mathbf{P}_2$) and orientation (${}^1 \mathbf{R}_2$) of the final e-e in the second platform's base frame. One can map ${}^1 \mathbf{R}_2$ to successive Z-Y-X Euler angles ψ, ϕ, θ as discussed in [23], pp. 54-56.

Using equation (2) the limb lengths of the first GS segment can be obtained when the mid-plate pose is determined. Knowing the final end-effector pose additionally, its translation to its local frame is possible using equations (3), (5). With local pose of the final end-effector in the upper GS frame, its six limb lengths can be once again derived using (2).

B. Jacobian of the Hybrid Manipulator

For a single GS segment, differentiating (1) relates the task-space twist $\mathbf{t} = [\dot{\mathbf{p}}^T \ \dot{\boldsymbol{\omega}}^T]^T$ to the joint-space velocity $\dot{\boldsymbol{\rho}}$ of its six limbs through the segment Jacobian $\mathbf{J}$ [24]:

$$\mathbf{t} = \begin{bmatrix} \mathbf{v} \\ \boldsymbol{\omega} \end{bmatrix} = \mathbf{J} \dot{\boldsymbol{\rho}}, \quad \mathbf{J} = \begin{bmatrix} \hat{\mathbf{s}}_1^T & (\mathbf{b}_1 \times \hat{\mathbf{s}}_1)^T \\ \vdots & \vdots \\ \hat{\mathbf{s}}_6^T & (\mathbf{b}_6 \times \hat{\mathbf{s}}_6)^T \end{bmatrix}_{6 \times 6}^{-1} \quad (6)$$

For the two-segment hybrid manipulator, the two segment Jacobians are combined via [24]:

$${}^G \mathbf{J}_2 = [\mathbf{V}_1 \ {}^G \mathbf{J}_1 \quad \mathbf{W}_1 \ {}^1 \mathbf{J}_2]_{6 \times 12} \quad (7)$$

where ${}^G \mathbf{J}_1$ is the first segment's Jacobian, already expressed in the global frame; ${}^1 \mathbf{J}_2$ is the second segment's Jacobian in its own base frame (the mid-plate); $\mathbf{V}_1$ and $\mathbf{W}_1$ introduced as below [24]:

$$\mathbf{V}_1 = \begin{bmatrix} 1_{3 \times 3} & -[{}^G \mathbf{p}_2]_{\times} \\ 0_{3 \times 3} & 1_{3 \times 3} \end{bmatrix}_{6 \times 6}, \quad \mathbf{W}_1 = \begin{bmatrix} {}^G \mathbf{R}_1 & 0_{3 \times 3} \\ 0_{3 \times 3} & {}^G \mathbf{R}_1 \end{bmatrix}_{6 \times 6} \quad (8)$$

where $[\dots]_{\times}$ denotes cross product matrix [23].

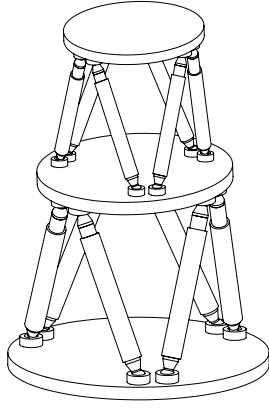


Fig. 5. Hybrid manipulator with two serially connected GS segments.

III. OPTIMIZATION PROBLEM

Redundancy of a two segment Gough-Stewart manipulator results in indefinitely many possible configurations for any unique specific point as the end-effector of the robot, which can be translated as infinite possible poses for the mid plate, i.e. the end-effector of the first GSP. Therefore, the redundancy problem can be formulated into an optimization problem where the design variables are the position (X, Y, Z) and orientation (θ, ϕ, ψ) of the mid-plate. There can be various objectives to this optimization problem, ranging from maximizing distance from singular configurations to minimizing cost functions like power consumption, with the latter considered in this study, for any specific end-effector pose and wrench.

Under a fixed operation time, the joint-space velocity $\dot{\mathbf{p}}$ is approximated by the joint-space displacement $\Delta \mathbf{l} = \mathbf{l} - \mathbf{l}_{\text{home}}$, measured from a reference (home) configuration of the twelve actuators. With a fixed end-effector wrench $\mathbf{w}$, the instantaneous power delivered to the load follows:

$$P_{\text{net}} = \mathbf{w}^T \mathbf{t} = \mathbf{w}^T \mathbf{J}_2 \Delta \mathbf{l} \quad (9)$$

Because the twelve-actuator system is kinematically redundant with respect to the six-dimensional task space, P_{net} admits internal actuator forces that do zero net work on the load while each individually consumes real power, so it is an unreliable measure of actual energy use. Taking the absolute value, $|P_{\text{net}}|$, resolves the sign ambiguity but not this cancellation, since the cancellation already happened inside the dot product. Instead, the gross power sums the magnitude of every actuator's individual contribution:

$$P_{\text{gross}} = \sum_{i=1}^{12} |\tau_i \Delta l_i|, \quad \boldsymbol{\tau} = \mathbf{J}_2^T \mathbf{w} \quad (10)$$

Since no cancellation between legs is possible in (10), P_{gross} is adopted as the optimization objective. It also weights each limb's contribution by its own actuator force, the six lower limbs – which carry the second segment's weight in addition to the end-effector wrench – are naturally penalized more than

the upper limbs, without requiring an artificial gain factor. The optimization problem is thus:

$$\begin{aligned} &\text{minimize} && f(\mathbf{X}) = P_{\text{gross}}(\mathbf{X}) + P_{\text{gross},1}(\mathbf{X}) \quad (11) \\ &\text{subject to} && \mathbf{X}^{\min} \leq \mathbf{X} \leq \mathbf{X}^{\max} \\ &\text{design vars} && \mathbf{X} = \{X, Y, Z, \theta, \phi, \psi\} \text{ of midplate} \end{aligned}$$

where $P_{\text{gross},1}$ is the analogous gross power the first segment's six actuators spend supporting the weight of the second segment, computed identically from ${}^G\mathbf{J}_1$ and a fixed weight-wrench acting on the mid-plate.

A. Genetic algorithm initialization

The chromosome in GA is composed of 6 genes which are the design variables considered in (11). The genetic algorithm is applied using MATLAB® `ga()` solver from the Global Optimization Toolbox, with its default operators for a bound-unconstrained problem. The population size, maximum number of generations, and crossover fraction were set to 1000, 500, and 0.8, respectively; all other operators were left at built-in defaults. The radius of the base, mid-plate, and end-effector plates were set to 0.5, 0.45, and 0.4 meters, respectively.

By default, `ga()` creates its initial population uniformly at random within the design-variable bounds, selects parents via stochastic uniform sampling weighted by scaled fitness, and preserves a small number of elite individuals unchanged into the next generation. The default crossover operator, scattered crossover, forms each gene of a child by drawing the corresponding gene from either parent according to a random binary mask. The default mutation operator, adaptive feasible, generates a random direction adaptive to the outcome of recent generations and takes a step of adaptive length along that direction, always producing a feasible point within the bounds. Since `ga()` solves a minimization problem directly, the objective $f(\mathbf{X})$ of (11) is used as-is. Because $f(\mathbf{X})$ already weights each limb's contribution by its own actuator force through (10), the higher load carried by the six lower limbs is accounted for directly by the physics of the problem, without requiring a hand-tuned gain.

IV. RESULTS

The genetic algorithm with all aforementioned settings has been implemented in MATLAB® with several case studies to confirm the practicality of proposed algorithm.

A. Main Symmetric Test Case

The most intuitable test case is the problem of finding the optimal mid-plate pose when the end-effector is fixed at the following point, under a fixed vertical wrench and a fixed weight for the second segment:

$$P_{\text{Goal}} = [0, 0, 3]^T, \quad R_{\text{Goal}} = \mathbf{I}_{3 \times 3} \quad (12)$$

$$\mathbf{w} = [0, 0, -50, 0, 0, 0]^T \text{ N, N}\cdot\text{m} \quad (13)$$

$$\mathbf{w}_{\text{seg2}} = [0, 0, -20, 0, 0, 0]^T \text{ N, N}\cdot\text{m} \quad (14)$$

TABLE I
HYPER-PARAMETERS CHOSEN FOR THE ALGORITHM SETTING AND THE RESULTS FOR THE MAIN SYMMETRIC TEST CASE.

	Setting					
	X	Y	Z	θ	ϕ	ψ
min	-1	-1	0.5	$-\pi/6$	$-\pi/6$	$-\pi/4$
max	1	1	2	$\pi/6$	$\pi/6$	$\pi/4$
goal	0	0	3	0	0	0
wrench	0	0	-50 (N)	0	0	0

	Result					
	l_1	l_2	l_3	l_4	l_5	l_6
seg. 1	1.3463	1.3463	1.3463	1.3463	1.3463	1.3463
seg. 2	1.82	1.82	1.82	1.82	1.82	1.82
mid	X	Y	Z	θ	ϕ	ψ
plate	0	0	1.25	0	0	0

This simple case configures the end-effector horizontally flat, at the origin of the X - Y plane at a Z coordinate of 3 units, with no tilt or twist, and under a purely vertical load. Due to the symmetry of the goal pose and the applied wrench about the Z axis, one can expect the optimal mid-plate to also be centered on this axis, with zero tilt.

Table I summarizes the hyper-parameters and results for the main test case. The reference (home) leg lengths l_{home} were computed at a mid-plate pose centered on the Z axis, untilted, at mid-travel height (Fig. 8(a)).

B. Validation of the Main Test Case

Because a genetic algorithm is a heuristic search over a non-convex objective, convergence to a candidate solution is not by itself a certificate of global optimality. The result reported in Table I was therefore validated through the following five checks:

- 1) *Convergence check*: Fig. 6 shows that the best-fitness curve was confirmed to flatten out via the function tolerance stopping criterion before the maximum generation count was reached.
- 2) *Multiple seeds*: the algorithm was re-run under several random seeds, all of which converged to equivalent mid-plate poses, indicating the result is not an artifact of one particular run.
- 3) *Local-solver polishing*: the genetic algorithm's result was used as the initial point for the gradient-based solver `fmincon`, which did not find a meaningfully lower gross power, confirming the genetic algorithm had already converged to at least a local optimum.
- 4) *First-order check*: a finite-difference estimate of the objective's gradient at the reported optimum was verified to vanish in every direction.
- 5) *Cross-check with an independent global algorithm*: the result was compared against optimizers described below.

C. Comparison with Alternative Global Optimizers

In addition to `particleswarm`, two further optimizers from MATLAB's Global Optimization Toolbox were used to cross-validate the result: `patternsearch`, a deterministic

TABLE II
MID-PLATE POSE OBTAINED BY GA AND ALTERNATIVE OPTIMIZERS.

Method	X	Y	Z	θ	ϕ	ψ
Test	GA	0	0	1.25	0	0
case 1	PSO	0	0	1.25	0	0
	PS	0	0	1.25	0	0
	SA	0	0	1.25	0	0
Test	GA	-0.649	1	0.5	-0.524	-0.573
case 2	PSO	-0.380	-0.289	1.283	-0.039	-0.242
						0.781
Test	GA	0	0	1.25	0	0
case 3	PSO	0	0	1.25	0	0
Test	GA	-0.024	0.399	1.279	-0.025	-0.014
case 4	PSO	-0.034	0.388	1.279	-0.02	-0.020
						-0.076
Test	GA	0.323	0.430	1.269	-0.095	0.217
case 5	PSO	0.316	0.416	1.272	-0.093	0.207
						0.352

direct-search method that does not require gradient information, and `simulannealbnd`, a simulated-annealing method that tolerates uphill moves and is therefore useful for detecting spurious optima artificially sitting at a design-variable bound. These three were chosen because, together with the genetic algorithm, they span population-based, deterministic-local, and stochastic-annealing search strategies, so agreement across all of them is a strong evidence of a genuine optimum.

The `particleswarm` was run with a swarm size of 200 for a maximum of 500 iterations. `patternsearch` and `simulannealbnd`, which refine a given starting point rather than using blind initial population, were both seeded at the genetic algorithm's result and run for a maximum of 2000 and 5000 iterations, respectively. Table II reports the resulting mid-plate pose for each method in different case studies.

D. Additional Test Cases

Beyond the symmetric case, several test cases were designed for which at least part of the optimal configuration could be predicted analytically, for further validation of the optimizer.

1) *Zero Wrench*: With $\mathbf{w} = \mathbf{0}$ and $\mathbf{w}_{\text{seg2}} = \mathbf{0}$, the actuator forces τ vanish identically regardless of pose, so $P_{\text{gross}} \equiv 0$ over the entire domain and every feasible pose is equally optimal. This case is primarily an implementation check: all optimizers are expected to report near-zero gross power while converging to different, arbitrary poses.

2) *Pure Torque About the Vertical Axis*: With the goal pose unchanged and $\mathbf{w} = [0, 0, 0, 0, 0, M_z]^T$, the applied wrench is itself invariant under the base's three-fold rotational symmetry about the Z axis. Since the only pose fixed by a nontrivial rotation about that axis is one centered on it, the optimal mid-plate is expected to again satisfy $X = Y = \theta = \phi = 0$, while Z and ψ are determined numerically.

3) *Off-Axis Goal Along the Base's Mirror Plane*: Shifting the goal along the base's single remaining mirror axis, $P_{\text{Goal}} = [0, 1, 3]^T$, breaks the three-fold symmetry but preserves the mirror symmetry about $X = 0$. The optimal mid-plate X coordinate is therefore expected to remain zero, while Y , Z , and the tilt angles become genuinely nonzero and only determinable numerically.

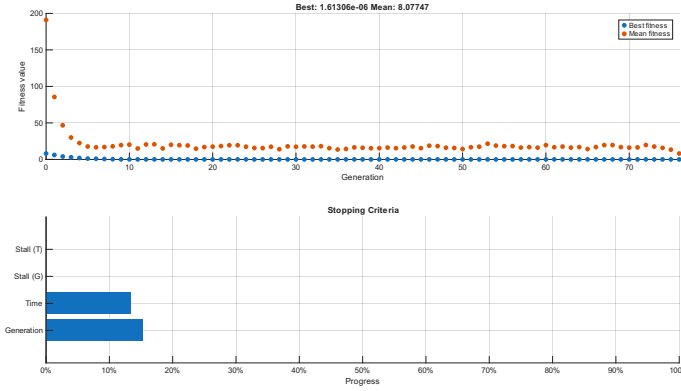


Fig. 6. Convergence behavior of MATLAB implementations for the main test case.

4) *Off-Axis Combined Force and Torque*: Superimposing a vertical force and a Z-axis torque, $\mathbf{w} = [0, 0, -50, 0, 0, M_z]^T$, on a goal with offset on all three axes, $P_{\text{Goal}} = [1, 1, 3]^T$, no longer preserves the symmetry argument and leans the mid-plate towards the goal. This case primarily checks that the gross power objective, which is not linear in the wrench, behaves consistently under superposition.

V. DISCUSSION

For the main test case described in the previous section, the IKP can be reduced to a 2D triangle problem in the X-Z or Y-Z plane. According to Fig. 8(b), any configuration except the one where the center of mid-plate lies on the line connecting the center of the base and the end-effector, would result in a triangle with a longer overall length. Thus, the optimal configuration is when the mid-plate is located on this line, which in the studied test case, is also the axis of symmetry of the problem. The red centerlines in Fig. 8(b) represent the overall distance (and consequently the average limb lengths) needed to reach the end-effector goal. Due to the triangle inequality theorem, in the depicted non-optimal configuration, the overall distance would be shorter if the mid-plate center was located on the dashed line.

The physical intuition behind the main test case can also be examined directly through the objective function's sensitivity to each design variable. Fig. 7 sweeps each of the six mid-plate pose variables individually, holding the remaining five at the optimized value, and plots the resulting gross and net power. All panels except Z show their minimum at zero. Therefore, the center of the mid-plate is expected to lie on the axis of symmetry where the X and Y coordinates are both zero. That is exactly achieved by the algorithm and shown in results.

The Z panel's minimum reflects the relative weighting between the tip wrench and the second segment's own weight, favoring a mid-plate height closer to whichever load dominates. In the main case study, the weight that the upper and lower segments carry are 50 and $50 + 20 = 70$ N, respectively. This is expected to result in a mid-plate Z coordinate closer to the base. Moreover, the ψ (yaw) panel is expected to show a minimum near zero, since a nonzero yaw twists every limb

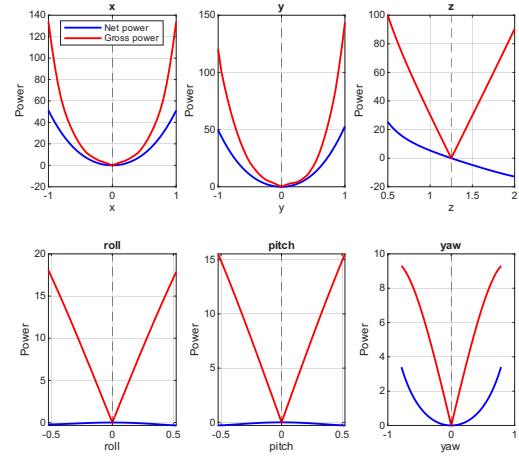


Fig. 7. Gross and net power consumption versus each design variable.

roughly evenly without contributing net displacement toward the goal. Table I and II both confirm the expectations.

The cross-validation against alternative optimizers in Table II depicts that all methods converge to proximately the same results. This behavior is observed for the other test cases, which satisfy the expectations explained for each one in previous section. Along with the other 4 checks mentioned in the previous section, this suggests a solid validation of the proposed genetic algorithm.

VI. CONCLUSION

In summary, the proposed algorithm is capable of solving the inverse kinematics problem of a hybrid redundant serial-parallel manipulator with two Gough-Stewart segments. The redundancy is resolved by optimizing the configuration of the robot to directly minimize the manipulator's gross power consumption, computed from the hybrid Jacobian. This formulation suggests optimal mid-plate pose for any fixed end-effector position and wrench. Since the actuator forces obtained from the Jacobian already reflect which limbs carry more load, the proposed approach allocates less share of movement to those limbs. The results are in line with the expected physics of each test case and are cross-validated against several independent global optimizers.

Future work can focus on extending the algorithm to more than two GSP segments, and considering additional objectives such as condition-number or manipulability measures leading to a multi-objective optimization problem.

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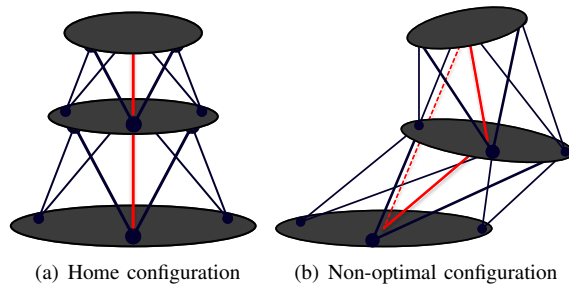


Fig. 8. Different configurations of a two stage SGSP.

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